COSTVISION · SOFTWARE SHOULD-COST REPORT

Range Rover L460 (PHEV)

Whole-vehicle software · PHEV · 52 software modules

Romania engineering hub · Tier-1 supplier · 200,000 vehicles a year × 6 years

Prepared 14 October 2026 · rate book CostVision built-in v2.0.0 (country book 2026-10) (reviewed 2026-10)

PROGRAMME TOTAL	PER VEHICLE, ALL-IN	ENGINEERING (NRE)	LIFECYCLE
£263.8M	**£220**	**£45.4M**	**£218.4M**
~ P16 of the simulated range	over 1,200,000 vehicles	4,806 person-months	to programme year 16

Read this first

- The figures are a deterministic should-cost model: every number comes from the module catalogue, the hub rate book and stated constants — no AI sets a price. The full arithmetic is in section 3 and was reconciled independently to the penny.
- 78.7 % of the total (£207.7M) rests on unsourced catalogue figures — chiefly cloud, map data, the OS licence placeholder and the module effort catalogue. Treat them as estimates to replace with supplier quotes.
- Accuracy against real programme costs has not been measured: no actual programme cost has been logged in the tool.
- The vehicle facts behind the scope were checked against the manufacturer's and press pages (section 2). The annual volume and programme life are the user's inputs, not published figures.

Programme	Range Rover L460 (PHEV)
Software scope	52 modules in 7 domains (section 5); excluded modules and catalogue gaps in section 12
Engineering hub	Romania — plus every hub in the rate book re-costed (section 8)
Countries	39 manufacturing countries mapped to their software hub (section 8)
Currency	£ — engine prices in £; figures shown in the report currency

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1 Executive summary

Range Rover L460 (PHEV): 52 software modules, 200,000 vehicles a year for 6 years (1,200,000 vehicles), engineered at the Romania hub by a tier-1 supplier.

Figure	Value	What it covers
Programme total	**£263.8M**	all engineering and every lifecycle cost to programme year 16
Per vehicle, all-in	£220	total ÷ 1,200,000 vehicles built
Engineering (NRE)	£45.4M	17.2 % — development, verification, integration, cyber, calibration, tools, homologation
Lifecycle	£218.4M	82.8 % — maintenance, licences and royalties, cloud
Royalty per vehicle built	£37.84	paid on every vehicle in the piece price
Running cost per vehicle-year	£22.25	map data (3 yr) + cloud (8 yr) per vehicle in service
Simulated range	£252.8M – £356.0M	P10–P90; P50 £299.6M; the headline sits at about P16
Hub range (same scope)	£227.3M – £376.4M	Egypt to USA — California (section 8)

What the figures say

- Lifecycle is 82.8 % of the total: cloud £103.2M, licences and royalties £97.4M, maintenance £17.8M. The running cost, more than the engineering, sets this programme's price.
- The largest domain is Infotainment, Connectivity & UX at £109.9M (41.7 %); the three largest modules — Navigation Stack £50.2M, IVI Operating System (Android Automotive / QNX) £28.7M, Cloud Backend Services £28.3M — are 40.7 % of the total.
- Engineering is 4,806 person-months at £5,390 per person-month loaded; moving the hub moves only the engineering and its maintenance — cloud, royalties and licences do not change with the hub.
- The total rests mostly on unsourced catalogue figures (78.7 %); the published-source share is 8.6 %. Replacing the largest unsourced lines with supplier quotes (section 10) is the fastest way to firm up the figure.

2 Programme definition

Every input the costing used. Defaults are the tool's stated assumptions; the basis of each is in section 10.

Input	Value	Basis
Vehicle / programme	Range Rover L460 (PHEV)	vehicle demo (facts below)
Drivetrain	PHEV	engine powertrain scope
Annual volume	200,000	user input
Programme life (production years)	6	user input
Engineering hub	Romania	hub rate book (appendix)
Development source	Tier-1 supplier	modelling assumption; efficiency factor unsourced, supplier margin sourced (section 10)
Senior engineer share	55 %	demo / user input
Engineering burden	employment +15 %, company +40 %, capital +0 %	section 10
Reuse (all modules unless set)	Medium	tool default Medium (× 0.60) — unsourced
Development window	54 months	Volkswagen's stated cycle (section 10)
Connected service per vehicle	8 years	median of current OEM terms
Map-data subscription per vehicle	3 years	OEM map-update terms
Security support after the last build	10 years	UN R155 post-production duty; share assumed
Discount rate	0 % (nominal money)	user input
Homologation	included	section 10

The vehicle

Range Rover (L460, 2022 on): JLR MLA-Flex platform, EVA 2.0 electrical architecture, Pivi Pro infotainment with software-over-the-air updates. PHEV: P460e / P550e, 38.2 kWh battery (MY24 on). Premium software from JLR-listed features: Dynamic Response Pro (48 V active roll control), all-wheel steering, Meridian Signature Sound, Cabin Air Purification Pro, head-up display (option), 3D surround camera and Remote Park Assist. Voice: Amazon Alexa. Digital key is not confirmed for the L460 and is out of scope. Tier-1 development is a modelling assumption, not JLR's sourcing.

Product facts the scope rests on (each checked on the page given)

Fact	Checked on
MLA-Flex platform; EVA 2.0 electrical architecture; Pivi Pro with software-over-the-air updates	https://media.landrover.com/introducing-new-range-rover-seamlessly-connected-convenience-and-luxury-every-turn
PHEV P460e / P550e, 38.2 kWh battery, up to 75 miles WLTP (from MY24)	https://www.fleetnews.co.uk/news/manufacturer-news/2023/05/17/range-rover-plug-in-hybrids-updated-with-more-power-and-range
Dynamic Response Pro (48 V electronic active roll control); all-wheel steering	https://media.landrover.com/introducing-new-range-rover-serene-capability-assurance-and-composure-and-road
Meridian Signature Sound System (35 loudspeakers)	https://meridian-audio.com/media/news-events/meridian-audio-brings-best-ever-audio-system-for-new-range-rover-to-mark-10-years-of-partnership/
Cabin Air Purification Pro (PM2.5 filtration)	https://www.landrover.in/range-rover/range-rover/explore.html
Head-up display (option), 3D surround camera, Remote Park Assist	https://media.landrover.com/introducing-new-range-rover-seamlessly-connected-convenience-and-luxury-every-turn
Amazon Alexa is the Pivi Pro voice assistant	https://www.landroverusa.com/our-story/news/amazon-alexa.html
No published software-cost figure for the L460 was found; no model-level volume is published (JLR reports Range Rover family wholesales)	https://media.jaguarlandrover.com/news/2024/04/jlr-full-year-sales-over-20-cent-year-year-reflecting-improved-production-and

3 How the cost is built

The seven steps the tool runs, with this programme's figures. An independent script recomputed every module and bucket from the same published tables and matched the tool to the penny at every hub (scripts/sw-review/rr-phev-trace.ts).

#	Step	Result	How
1	**Scope**	52 modules	7 software domains on a PHEV vehicle; nominal effort 2,472 person-months (fresh code, QM, medium complexity).
2	**Development effort**	2,779 PM	Reuse, complexity and the ISO 26262 ASIL uplift applied to each module: 2,472 -> 2,779 PM. 18 modules at ASIL C/D (lit).
3	**Verification & integration**	+2,027 PM	SIL / MIL / HIL testing, integration, ISO/SAE 21434 cybersecurity and calibration: 4,806 PM of engineering in all. 20 modules at CAL3/4 (lit).
4	**Engineering rate**	£5,390 / PM	Romania: 55 % senior blend of the hub's mid and senior rates; mid £3,359 / senior £3,359 per person-month (salary + statutory on-cost ÷ productive hours × 152 h), × development source price × 1.61 engineering burden.
5	**Engineering cost (NRE)**	£45.4M	Effort × rate = £25.9M; tools £15.8M, homologation £3.7M. Largest domain: Infotainment, Connectivity & UX £109.9M (all costs).
6	**Lifecycle**	£218.4M	Maintenance £17.8M, cloud £103.2M, licences and royalties £97.4M — paid over the 6 production years and the service years after the last build (to programme year 16).
7	**Per vehicle**	£220	NRE £45.4M over 1,200,000 vehicles (6-year recovery) + lifecycle over 1,200,000 vehicles. Programme total £263.8M.

Formulas

- Loaded rate = hub mid / senior blend × development-source price factor × engineering burden.
- Development person-months = catalogue person-months × reuse × ASIL factor; costed across requirements 12 %, architecture 14 %, algorithms 22 % × complexity, implementation 37 % × (1 + 0.15 × (complexity - 1)), safety 15 % (reuse floored by ASIL).
- Testing, integration, cybersecurity (by the ISO/SAE 21434 CAL) and calibration are fractions of development; tools = engineer-years × the seat price.
- Maintenance = (development + testing + integration) × the module's % a year — full for 3 years after SOP, then 50 %, then the post-production security share after the last build.
- Royalties = vehicles × £ per vehicle built + vehicle-years × £ per vehicle-year; cloud = connected vehicle-years × £ per connected vehicle-year.

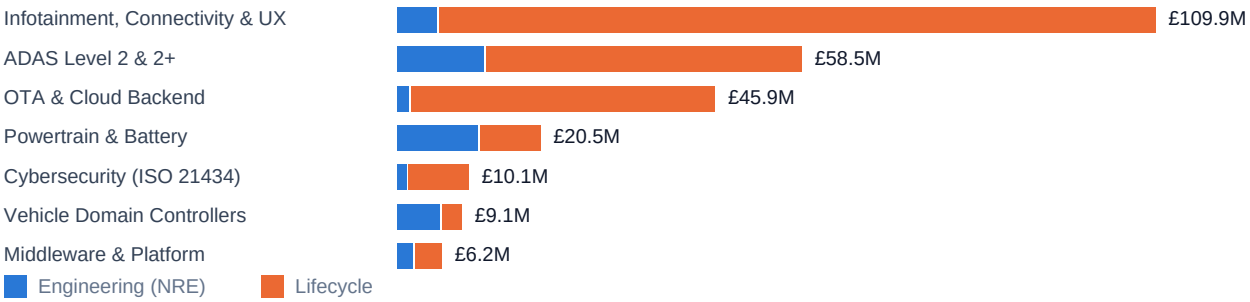
4 Cost build-up

Every bucket, the two subtotals and the total. The check line confirms the buckets add up.

Bucket	£M	Share
Development engineering	£15.0M	5.7 %
Testing and validation (SIL / MIL / HIL)	£6.2M	2.3 %
Integration	£2.8M	1.0 %
Cybersecurity engineering	£0.8M	0.3 %
Calibration	£1.2M	0.5 %
Development tools	£15.8M	6.0 %
Homologation (R155 / R156 / ISO 26262)	£3.7M	1.4 %
Engineering (NRE) subtotal	**£45.4M**	**17.2 %**
Maintenance and security support	£17.8M	6.7 %
Licences and royalties	£97.4M	36.9 %
Cloud	£103.2M	39.1 %
Lifecycle subtotal	**£218.4M**	**82.8 %**
Programme total	**£263.8M**	**100.0 %**

Check, exact: NRE £45,376,243 + lifecycle £218,419,311 = £263,795,554; the programme total is £263,795,554 (they agree).

By software domain



Domain	Modules	NRE	Lifecycle	Total	Share
Infotainment, Connectivity & UX	8	£5.8M	£104.1M	£109.9M	41.7 %
ADAS Level 2 & 2+	9	£12.6M	£45.9M	£58.5M	22.2 %
OTA & Cloud Backend	4	£1.7M	£44.2M	£45.9M	17.4 %
Powertrain & Battery	13	£11.8M	£8.8M	£20.5M	7.8 %
Cybersecurity (ISO 21434)	5	£1.3M	£8.8M	£10.1M	3.8 %
Vehicle Domain Controllers	7	£6.2M	£2.9M	£9.1M	3.4 %
Middleware & Platform	6	£2.3M	£3.8M	£6.2M	2.3 %

Domain totals are module costs; homologation (programme-level) sits outside the domains.

5 Every software module

All 52 modules in scope, grouped by domain. PM = costed engineering person-months (development, verification, integration, cyber, calibration). NRE includes the module's tools; lifecycle is its maintenance, licences, royalties and cloud.

Module	ASIL	Complexity	Reuse	CAL	PM	NRE	Lifecycle	Total
Infotainment, Connectivity & UX								
Navigation Stack	QM	High	Medium	—	44	£0.4M	£49.9M	£50.2M
IVI Operating System (Android Automotive / QNX)	QM	Very High	Medium	CAL4	248	£2.2M	£26.5M	£28.7M
Telematics Control Unit (TCU) Software	B	High	Medium	CAL4	51	£0.4M	£20.4M	£20.8M
Vehicle Voice Assistant	QM	Very High	Medium	CAL4	100	£0.9M	£5.1M	£5.9M
App Framework & HMI Layer	QM	Very High	Medium	—	112	£1.0M	£1.0M	£2.0M
Bluetooth / WiFi / 5G Connectivity Stack	QM	High	Medium	CAL4	34	£0.3M	£0.7M	£1.0M
Premium Audio DSP & Active Noise Control	QM	Very High	Medium	—	38	£0.3M	£0.3M	£0.7M
AR Head-Up Display & AR Navigation	A	High	Medium	—	40	£0.3M	£0.2M	£0.6M
ADAS Level 2 & 2+								
Camera Perception Stack	B	Very High	Medium	CAL3	304	£2.6M	£20.9M	£23.5M
Sensor Fusion Engine	C	Very High	Medium	—	246	£2.1M	£5.9M	£8.0M
Highway Assist / Traffic Jam Assist	C	Very High	Medium	—	196	£1.7M	£6.1M	£7.8M
Path Planning Algorithms	C	Very High	Medium	—	202	£1.8M	£4.5M	£6.3M
Driver Monitoring Software (DMS)	B	Very High	Medium	CAL2	121	£1.0M	£5.0M	£6.0M
Radar Processing Stack	B	Very High	Medium	—	127	£1.1M	£2.4M	£3.5M
Control Algorithms (ACC / LKA / AEB)	D	High	Medium	—	152	£1.3M	£0.7M	£2.1M
Automated Parking & Surround-View	B	High	Medium	—	81	£0.7M	£0.3M	£1.0M
Ultrasonic Processing Software	B	Medium	Medium	—	19	£0.2M	£0.1M	£0.3M
OTA & Cloud Backend								
Cloud Backend Services	QM	Very High	Medium	CAL4	96	£0.8M	£27.5M	£28.3M
Data Pipeline & Telemetry	QM	High	Medium	CAL3	34	£0.3M	£7.5M	£7.8M
OTA Update Manager (Vehicle-side)	B	High	Medium	CAL4	52	£0.5M	£4.4M	£4.9M
Fleet Management & Analytics Software	QM	Medium	Medium	—	17	£0.1M	£4.7M	£4.9M
Powertrain & Battery								
SOC/SOH/SOE Estimation Models	C	High	Medium	—	81	£0.7M	£3.1M	£3.8M
Hybrid Supervisory Control & Energy Management	C	Very High	Medium	—	210	£1.8M	£0.9M	£2.7M
BMS Core Software	D	High	Medium	CAL4	211	£1.8M	£0.8M	£2.6M
EDU (Electric Drive Unit) Control Software	D	High	Medium	CAL4	159	£1.4M	£0.6M	£2.0M
Engine Management Software (ECU)	B	Very High	Medium	CAL3	150	£1.3M	£0.6M	£1.9M
Inverter Control Algorithms	D	Very High	Medium	—	125	£1.1M	£0.4M	£1.5M
Motor Control (FOC/DTC/SVPWM)	D	Very High	Medium	—	118	£1.0M	£0.4M	£1.4M
Transmission Control Software (TCU)	B	High	Medium	—	103	£0.9M	£0.4M	£1.3M
Fast-Charging Control Software	C	Medium	Medium	CAL3	41	£0.4M	£0.9M	£1.2M
Battery Thermal Management Software	B	High	Medium	—	48	£0.4M	£0.2M	£0.6M
Regenerative Braking Software	C	High	Medium	—	38	£0.3M	£0.1M	£0.5M
Cell Balancing Algorithms	C	High	Medium	—	37	£0.3M	£0.1M	£0.5M
Emissions After-treatment & OBD	QM	High	Medium	—	34	£0.3M	£0.1M	£0.4M
Cybersecurity (ISO 21434)								
Secure OTA Software Update Framework	B	High	Medium	CAL4	35	£0.3M	£3.0M	£3.3M
Intrusion Detection System (IDS)	QM	High	Medium	CAL3	35	£0.3M	£2.4M	£2.7M

Module	ASIL	Complexity	Reuse	CAL	PM	NRE	Lifecycle	Total
Key Management System	B	High	Medium	CAL4	26	£0.2M	£2.0M	£2.2M
Secure Boot & Root of Trust	B	High	Medium	CAL3	36	£0.3M	£1.2M	£1.5M
Encryption Modules & Crypto Stack	B	Medium	Medium	CAL3	21	£0.2M	£0.2M	£0.3M
Vehicle Domain Controllers								
Vehicle Motion Management (VMM)	C	Very High	Medium	—	210	£1.8M	£0.9M	£2.7M
Zonal Architecture Software	B	Very High	Medium	CAL4	145	£1.3M	£0.6M	£1.9M
Chassis Control Software	C	High	Medium	—	118	£1.0M	£0.4M	£1.5M
Predictive Active Suspension & Body Control	C	Very High	Medium	—	109	£0.9M	£0.4M	£1.4M
Body Control Software	B	Medium	Medium	—	58	£0.5M	£0.2M	£0.7M
Gateway ECU Software	B	High	Medium	CAL4	47	£0.4M	£0.2M	£0.6M
Climate & Thermal Comfort Control	A	Medium	Medium	—	28	£0.2M	£0.1M	£0.3M
Middleware & Platform								
AUTOSAR Adaptive Platform	B	High	Medium	CAL3	105	£0.9M	£1.7M	£2.6M
AUTOSAR Classic BSW & Integration	C	High	Medium	—	82	£0.7M	£1.0M	£1.7M
Diagnostics (UDS / OBD-II)	A	Medium	Medium	—	26	£0.2M	£0.7M	£0.9M
CAN / LIN / FlexRay / Ethernet Communication Stacks	B	Medium	Medium	—	25	£0.2M	£0.2M	£0.4M
RTOS & OS Porting	C	Medium	Medium	—	18	£0.2M	£0.2M	£0.4M
Time Synchronisation & Network Management	A	Low	Medium	—	13	£0.1M	£0.1M	£0.2M
All modules					**4,806**	**£41.7M**	**£218.4M**	**£260.1M**

Check, exact: module sum £260,140,153 + homologation £3,655,401 = programme total £263,795,554.

6 Lifecycle and unit economics

The figures purchasing compares, and every per-vehicle charge with the basis it was priced on.

Figure	Value	Volume	Programme £
Engineering, one-off (NRE)	£45.4M	—	£45.4M
Royalty per vehicle built	£37.84	1,200,000 vehicles	£45.4M
Map data per vehicle-year	£11.50	3,600,000 vehicle-years (3 yr each)	£41.4M
Cloud per connected vehicle-year	£10.75	9,600,000 vehicle-years (8 yr each)	£103.2M
Fixed per production year (maintenance, IP licences, platform) — average	£4.7M	× 6 years	£28.4M
Programme total			**£263.8M**

Royalties per vehicle — each priced on its own basis

Module	Programme £	Grade	Basis
Navigation Stack	£41.4M	Unsourced catalogue	Map data £8–15 / vehicle / yr — midpoint £11.50, an unsourced estimate: neither HERE nor TomTom publishes a per-car price (TomTom FY25: automotive revenue €322M, backlog €2.4bn, no vehicle count), paid for the subscription years after each vehicle is built (default 3 — OEM map-update terms).
IVI Operating System (Android Automotive / QNX)	£22.7M	Unsourced catalogue	Unsourced placeholder ~$25 / vehicle for Google Automotive Services (GAS) or a QNX runtime licence — neither price is public: Google does not publish a GAS fee, and BlackBerry licenses QNX per unit but publishes no per-vehicle royalty (FY26 QNX revenue US$268M, royalty backlog ~US$950M, 275M+ vehicles — SEC 10-K). Android Automotive OS itself is open source with no licence fee. Replace with your supplier quote.
Telematics Control Unit (TCU) Software	£18.1M	Published source	Avanci 4G vehicle licence (cellular standard-essential patents) US$20 per vehicle, paid once by the car maker — avanci.com published rate (5G US$32, 3G US$9, eCall only US$3), at the app's USD_PER_GBP. Set to 0 if your TCU supplier's price already carries it.
Vehicle Voice Assistant	£4.6M	Published source	Cerence average price per unit US$5.09 (TTM to Q2 FY2026; US$5.05 at Q4 FY2025; includes connected services) at the app's USD_PER_GBP — Cerence earnings decks filed with the SEC (EX-99), read via search results.

Embedded IP licences (£ a year per module, over the production years)

Module	Programme £	Module	Programme £
Camera Perception Stack	£1.0M	AUTOSAR Classic BSW & Integration	£0.8M
Cloud Backend Services	£0.8M	AUTOSAR Adaptive Platform	£0.7M
Driver Monitoring Software (DMS)	£0.5M	App Framework & HMI Layer	£0.5M
Highway Assist / Traffic Jam Assist	£0.4M	Data Pipeline & Telemetry	£0.4M
Sensor Fusion Engine	£0.3M	OTA Update Manager (Vehicle-side)	£0.3M
Path Planning Algorithms	£0.3M	Intrusion Detection System (IDS)	£0.3M
Fleet Management & Analytics Software	£0.3M	Premium Audio DSP & Active Noise Control	£0.3M
Radar Processing Stack	£0.2M	Bluetooth / WiFi / 5G Connectivity Stack	£0.2M
RTOS & OS Porting	£0.2M	Secure OTA Software Update Framework	£0.2M
Vehicle Motion Management (VMM)	£0.2M	Key Management System	£0.2M
Hybrid Supervisory Control & Energy Management	£0.2M	SOC/SOH/SOE Estimation Models	£0.1M
EDU (Electric Drive Unit) Control Software	£0.1M	Control Algorithms (ACC / LKA / AEB)	£0.1M
Zonal Architecture Software	£0.1M	CAN / LIN / FlexRay / Ethernet Communication Stacks	£0.1M
Secure Boot & Root of Trust	£0.1M	Engine Management Software (ECU)	£0.1M
BMS Core Software	£0.1M	Telematics Control Unit (TCU) Software	£0.1M
Chassis Control Software	£0.1M	AR Head-Up Display & AR Navigation	£0.1M
Transmission Control Software (TCU)	£0.1M	Diagnostics (UDS / OBD-II)	£0.1M

Module	Programme £	Module	Programme £
Encryption Modules & Crypto Stack	£0.1M	Predictive Active Suspension & Body Control	£0.1M
Fast-Charging Control Software	£0.1M	Inverter Control Algorithms	£0.1M
Motor Control (FOC/DTC/SVPWM)	£0.1M	Gateway ECU Software	£0.1M
Automated Parking & Surround-View	£0.1M	Body Control Software	£0.1M
Ultrasonic Processing Software	£0.0M	Cell Balancing Algorithms	£0.0M
Battery Thermal Management Software	£0.0M	Regenerative Braking Software	£0.0M
Time Synchronisation & Network Management	£0.0M	Climate & Thermal Comfort Control	£0.0M
Emissions After-treatment & OBD	£0.0M		

IP licences £10.6M in all — the module catalogue's £ a year, unsourced (no licence prices are published for these stacks).

Cloud per module

Module	Programme £	Module	Programme £
Cloud Backend Services	£26.2M	Camera Perception Stack	£18.5M
Navigation Stack	£8.3M	Data Pipeline & Telemetry	£7.0M
Highway Assist / Traffic Jam Assist	£4.8M	Sensor Fusion Engine	£4.4M
Fleet Management & Analytics Software	£4.4M	Driver Monitoring Software (DMS)	£3.9M
OTA Update Manager (Vehicle-side)	£3.9M	Path Planning Algorithms	£3.3M
SOC/SOH/SOE Estimation Models	£2.6M	IVI Operating System (Android Automotive / QNX)	£2.6M
Secure OTA Software Update Framework	£2.6M	Telematics Control Unit (TCU) Software	£2.0M
Intrusion Detection System (IDS)	£2.0M	Radar Processing Stack	£1.7M
Key Management System	£1.7M	Secure Boot & Root of Trust	£1.0M
Fast-Charging Control Software	£0.7M	AUTOSAR Adaptive Platform	£0.7M
Diagnostics (UDS / OBD-II)	£0.5M	Bluetooth / WiFi / 5G Connectivity Stack	£0.4M

Cloud £103.2M in all. Basis (every module): Module cloud £ / yr ÷ reference fleet — an unsourced estimate (~ £10.8 per connected vehicle-year with every module in scope). No OEM publishes a back-end cost; the only list price, AWS IoT FleetWise (US$0.60 / 0.45 per vehicle-month, messages and storage extra; closed to new customers 30 Apr 2026), is US$5.4–7.2 per vehicle-year for data collection alone. Each connected vehicle served 8 years from build — the median of current OEM terms: GM OnStar Basics 8 yr (MY2025+), Tesla Standard Connectivity 8 yr (ordered after 20 Jul 2022), Toyota up to 10 yr, Mercedes-Benz mbrace Connect 5 yr (US); OEM pages, re-checked 14 Oct 2026. A 2G / 3G / 4G shutdown can end it sooner.

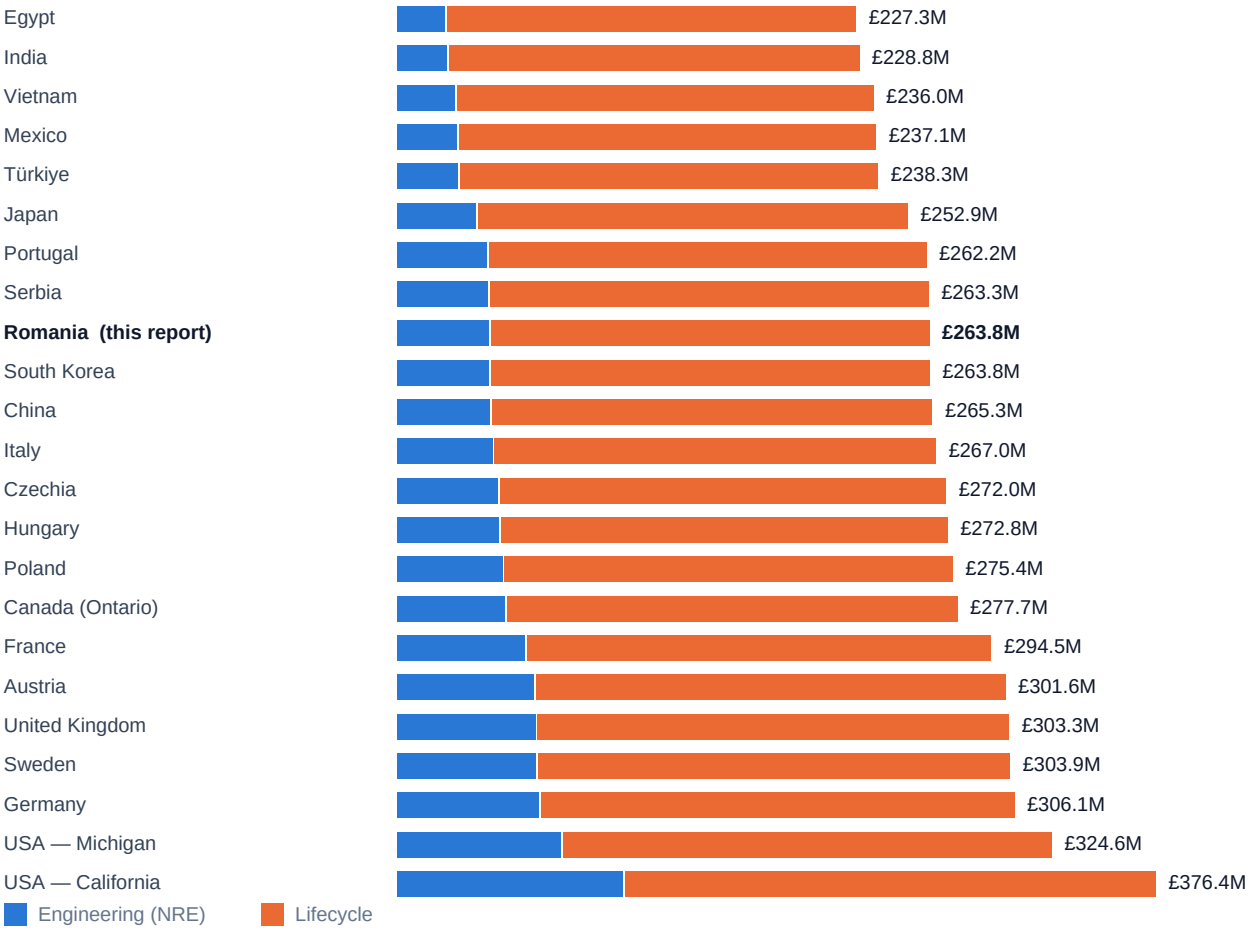
7 Spend by year

Nominal £ (no discounting). Engineering over 54 development months by phase share; maintenance in full for 3 years after SOP then a tail; royalties per vehicle built and for 3 subscription years; cloud for 8 years per connected vehicle, so service years run past the last build year.

Year	Engineering	Maintenance	Licences	Royalties	Cloud	Total	Cumulative
Dev 1	£10.2M	£0.0M	£0.0M	£0.0M	£0.0M	£10.2M	£10.2M
Dev 2	£13.0M	£0.0M	£0.0M	£0.0M	£0.0M	£13.0M	£23.1M
Dev 3	£11.1M	£0.0M	£0.0M	£0.0M	£0.0M	£11.1M	£34.2M
Dev 4	£7.3M	£0.0M	£0.0M	£0.0M	£0.0M	£7.3M	£41.5M
Dev 5	£3.9M	£0.0M	£0.0M	£0.0M	£0.0M	£3.9M	£45.4M
Year 1	£0.0M	£3.1M	£1.8M	£9.9M	£2.2M	£16.9M	£62.3M
Year 2	£0.0M	£3.1M	£1.8M	£12.2M	£4.3M	£21.3M	£83.6M
Year 3	£0.0M	£3.1M	£1.8M	£14.5M	£6.5M	£25.8M	£109.4M
Year 4	£0.0M	£1.5M	£1.8M	£14.5M	£8.6M	£26.4M	£135.8M
Year 5	£0.0M	£1.5M	£1.8M	£14.5M	£10.7M	£28.5M	£164.3M
Year 6	£0.0M	£1.5M	£1.8M	£14.5M	£12.9M	£30.7M	£195.0M
Service 1	£0.0M	£0.4M	£0.0M	£4.6M	£12.9M	£17.9M	£212.9M
Service 2	£0.0M	£0.4M	£0.0M	£2.3M	£12.9M	£15.6M	£228.5M
Service 3	£0.0M	£0.4M	£0.0M	£0.0M	£10.7M	£11.1M	£239.6M
Service 4	£0.0M	£0.4M	£0.0M	£0.0M	£8.6M	£9.0M	£248.6M
Service 5	£0.0M	£0.4M	£0.0M	£0.0M	£6.5M	£6.8M	£255.4M
Service 6	£0.0M	£0.4M	£0.0M	£0.0M	£4.3M	£4.7M	£260.1M
Service 7	£0.0M	£0.4M	£0.0M	£0.0M	£2.2M	£2.5M	£262.6M
Service 8	£0.0M	£0.4M	£0.0M	£0.0M	£0.0M	£0.4M	£263.0M
Service 9	£0.0M	£0.4M	£0.0M	£0.0M	£0.0M	£0.4M	£263.4M
Service 10	£0.0M	£0.4M	£0.0M	£0.0M	£0.0M	£0.4M	£263.8M
Total	**£45.4M**	**£17.8M**	**£10.6M**	**£86.8M**	**£103.2M**	**£263.8M**	

8 Every engineering hub and every country

The same 52-module scope re-costed at each of the 23 hubs in the rate book, each at its own salaries, employer on-costs and working hours (no UK multiplier). Only engineering and its maintenance move with the hub.



Hub	Rate confidence	Loaded £/PM	NRE	Lifecycle	Total	Per vehicle	v this report
Egypt	Low	£882	£23.7M	£203.5M	£227.3M	£189	-£36.5M
India	Low	£1,072	£24.6M	£204.2M	£228.8M	£191	-£35.0M
Vietnam	Low	£1,955	£28.9M	£207.1M	£236.0M	£197	-£27.8M
Mexico	Low	£2,101	£29.6M	£207.6M	£237.1M	£198	-£26.7M
Türkiye	Low	£2,240	£30.2M	£208.0M	£238.3M	£199	-£25.5M
Japan	Medium	£4,051	£38.9M	£214.0M	£252.9M	£211	-£10.8M
Portugal	Low	£5,198	£44.5M	£217.8M	£262.2M	£219	-£1.6M
Serbia	Medium	£5,333	£45.1M	£218.2M	£263.3M	£219	-£0.5M
Romania	Low	£5,390	£45.4M	£218.4M	**£263.8M**	£220	—
South Korea	Medium	£5,396	£45.4M	£218.4M	£263.8M	£220	+£0.1M
China	Low	£5,573	£46.3M	£219.0M	£265.3M	£221	+£1.5M
Italy	Low	£5,788	£47.3M	£219.7M	£267.0M	£223	+£3.2M
Czechia	Medium	£6,399	£50.2M	£221.8M	£272.0M	£227	+£8.2M
Hungary	Low	£6,500	£50.7M	£222.1M	£272.8M	£227	+£9.0M

Hub	Rate confidence	Loaded £/PM	NRE	Lifecycle	Total	Per vehicle	v this report
Poland	Low	£6,824	£52.3M	£223.2M	£275.4M	£230	+£11.6M
Canada (Ontario)	Medium	£7,108	£53.6M	£224.1M	£277.7M	£231	+£13.9M
France	Medium	£9,181	£63.6M	£230.9M	£294.5M	£245	+£30.7M
Austria	Low	£10,058	£67.8M	£233.8M	£301.6M	£251	+£37.8M
United Kingdom	Medium	£10,267	£68.8M	£234.5M	£303.3M	£253	+£39.5M
Sweden	Medium	£10,338	£69.2M	£234.7M	£303.9M	£253	+£40.1M
Germany	Medium	£10,605	£70.4M	£235.6M	£306.1M	£255	+£42.3M
USA — Michigan	Medium	£12,890	£81.4M	£243.2M	£324.6M	£270	+£60.8M
USA — California	Medium	£19,284	£112.2M	£264.3M	£376.4M	£314	+£112.6M

Every manufacturing country and its software hub (39)

Software is costed where it is engineered. A country with its own assessed hub uses it; any other uses the nearest assessed hub, for the reason given.

Country	Software hub	Programme total at that hub	Why this hub
Austria	Austria	£301.6M	its own assessed hub
Belgium	Germany	£306.1M	Western Europe — Germany is the nearest assessed hub
Brazil	Mexico	£237.1M	Latin America — Mexico (Guadalajara) is the assessed hub
Bulgaria	Romania	£263.8M	South-East Europe — Romania is the nearest assessed hub
Canada	Canada (Ontario)	£277.7M	its own assessed hub
China	China	£265.3M	its own assessed hub
Czech Republic	Czechia	£272.0M	its own assessed hub
Egypt	Egypt	£227.3M	its own assessed hub
France	France	£294.5M	its own assessed hub
Germany	Germany	£306.1M	its own assessed hub
Hungary	Hungary	£272.8M	its own assessed hub
India	India	£228.8M	its own assessed hub
Indonesia	Vietnam	£236.0M	South-East Asia — Vietnam is the assessed hub
Italy	Italy	£267.0M	its own assessed hub
Japan	Japan	£252.9M	its own assessed hub
Lithuania	Poland	£275.4M	Baltic — Poland is the nearest assessed hub
Malaysia	Vietnam	£236.0M	South-East Asia — Vietnam is the assessed hub
Mexico	Mexico	£237.1M	its own assessed hub
Morocco	Egypt	£227.3M	North Africa — Egypt (Valeo Cairo) is the assessed North-African hub
Netherlands	Germany	£306.1M	Western Europe — Germany is the nearest assessed hub
Philippines	Vietnam	£236.0M	South-East Asia — Vietnam is the assessed hub
Poland	Poland	£275.4M	its own assessed hub
Portugal	Portugal	£262.2M	its own assessed hub
Romania	Romania	£263.8M	its own assessed hub
Serbia	Serbia	£263.3M	its own assessed hub
Singapore	Japan	£252.9M	high-cost Asian engineering — Japan is the nearest assessed hub
Slovakia	Czechia	£272.0M	Central Europe — Czechia is the nearest assessed hub

Country	Software hub	Programme total at that hub	Why this hub
Slovenia	Austria	£301.6M	Central Europe — Austria is the nearest assessed hub
South Africa	India	£228.8M	no assessed African hub south of Egypt — India, the largest offshore automotive software base
South Korea	South Korea	£263.8M	its own assessed hub
Spain	Portugal	£262.2M	Iberia — the research found no Spanish embedded-automotive salary data; Portugal (Bosch Braga) is the assessed Iberian hub
Sweden	Sweden	£303.9M	its own assessed hub
Taiwan	China	£265.3M	Greater China — China is the assessed hub
Thailand	Vietnam	£236.0M	South-East Asia — Vietnam is the assessed hub
Tunisia	Egypt	£227.3M	North Africa — Egypt (Valeo Cairo) is the assessed North-African hub
Turkey	Türkiye	£238.3M	its own assessed hub
United Kingdom	United Kingdom	£303.3M	its own assessed hub
United States	USA — Michigan	£324.6M	United States — Michigan (Detroit / Ann Arbor OEM software centres); choose California for Bay Area teams
Vietnam	Vietnam	£236.0M	its own assessed hub

9 Risk and sensitivity

1,000 seeded trials; each costed line is drawn from a triangular range set by its evidence grade, with a 55 % programme-wide correlation. The headline adds each line's most likely value; because most money rests on unsourced lines whose risk runs upwards, it sits at about P16, not at the median.

Measure	Programme total	Per vehicle
P10	£252.8M	£211
P50 (median)	**£299.6M**	**£250**
P90	£356.0M	£297
Mean	£302.5M	£252
Headline (~ P16)	£263.8M	£220

What moves the total — one driver at a time, everything else held

Driver (low case vs high case)	Low	Base	High	Swing
Programme life (4 vs 9 years)	£193.8M	£263.8M	£368.8M	£174.9M
Engineering hub (Egypt vs USA — California)	£227.3M	£263.8M	£376.4M	£149.2M
Connected-service life (5 vs 15 years)	£225.1M	£263.8M	£354.1M	£129.0M
Effort catalogue (× 0.5 vs × 2)	£234.0M	£263.8M	£323.3M	£89.3M
Map subscription (1 vs 7 years)	£236.2M	£263.8M	£319.0M	£82.8M
Reuse level (Heavy vs Fresh)	£240.2M	£263.8M	£303.5M	£63.3M
Complexity (all modules: Medium vs Very High)	£248.5M	£263.8M	£270.4M	£21.9M
ASIL level (all modules: B vs D)	£263.2M	£263.8M	£281.4M	£18.1M
Company burden (20 % vs 60 %)	£257.6M	£263.8M	£270.0M	£12.5M
Security support after the last build (0 vs 15 years)	£259.9M	£263.8M	£265.7M	£5.8M
Production volume (300k vs 100k a year, per vehicle)	£199	£220	£281	£82

10 Evidence and sources

How well each part of the total is supported, the basis of every stated assumption, and the rate build-up.

Evidence grade	In the total	Share	Simulated range (low · high)
Your figures	£0.0M	0.0 %	× 0.9 · × 1.15
Calibrated to your actuals	£0.0M	0.0 %	× 0.85 · × 1.25
Published source	£22.7M	8.6 %	× 0.85 · × 1.2
Labelled estimate	£33.4M	12.7 %	× 0.7 · × 1.6
Unsourced catalogue	£207.7M	78.7 %	× 0.6 · × 1.9

Largest unsourced amounts — replace these first

What	In the total
Cloud (per connected vehicle-year)	£103.2M
Royalties	£64.1M
Engineering effort (module catalogue)	£25.9M
IP licences (module catalogue)	£10.6M
Security support after the last build (share assumed)	£3.9M

Your figures — quotes and modules you entered, with their sources

None yet. Enter supplier quotes on Setup (Supplier quotes & licence register) and modules the catalogue lacks on Scope (Your modules) — each replaces an unsourced figure.

Calibration — which modules to measure first

This costing is uncalibrated. £25.9M of engineering rests on the catalogue's person-months. Log what these modules actually took (download the template on Rates & method -> Calibrate to your actuals); the top 5 cover 25.4 % of it.

Module	Person-months	Engineering £	Share	Cumulative
Camera Perception Stack	304	£1.6M	6.3 %	6.3 %
IVI Operating System (Android Automotive / QNX)	248	£1.3M	5.2 %	11.5 %
Sensor Fusion Engine	246	£1.3M	5.1 %	16.6 %
BMS Core Software	211	£1.1M	4.4 %	21.0 %
Vehicle Motion Management (VMM)	210	£1.1M	4.4 %	25.4 %
Hybrid Supervisory Control & Energy Management	210	£1.1M	4.4 %	29.7 %
Path Planning Algorithms	202	£1.1M	4.2 %	33.9 %
Highway Assist / Traffic Jam Assist	196	£1.1M	4.1 %	38.0 %
EDU (Electric Drive Unit) Control Software	159	£0.9M	3.3 %	41.3 %
Control Algorithms (ACC / LKA / AEB)	152	£0.8M	3.2 %	44.5 %
Engine Management Software (ECU)	150	£0.8M	3.1 %	47.6 %
Zonal Architecture Software	145	£0.8M	3.0 %	50.6 %

Stated assumptions and their basis

Assumption	Basis

Assumption	Basis
Connected service per vehicle	Each connected vehicle served 8 years from build — the median of current OEM terms: GM OnStar Basics 8 yr (MY2025+), Tesla Standard Connectivity 8 yr (ordered after 20 Jul 2022), Toyota up to 10 yr, Mercedes-Benz mbrace Connect 5 yr (US); OEM pages, re-checked 14 Oct 2026. A 2G / 3G / 4G shutdown can end it sooner.
Map-data subscription	3 years of map updates per vehicle — Nissan, Subaru, Hyundai, Honda (2021–24) include 3; some brands 5 or lifetime (OEM pages, read from search results).
Cloud £ per connected vehicle-year	Module cloud £ / yr ÷ reference fleet — an unsourced estimate (~ £10.8 per connected vehicle-year with every module in scope). No OEM publishes a back-end cost; the only list price, AWS IoT FleetWise (US$0.60 / 0.45 per vehicle-month, messages and storage extra; closed to new customers 30 Apr 2026), is US$5.4–7.2 per vehicle-year for data collection alone.
Development window	54 months feasibility -> SOP: Volkswagen's stated development cycle (IAA 2023; target 36 months). GM "around four years" (2022); Chinese OEMs ~18–24 months. Set your programme's own.
Development tools	CostVision estimate £39,500 per engineer-seat-year — the module catalogue's tool £ at the default programme ÷ its engineer-years; no public seat price (Vector CANoe is quote-only).
Maintenance profile	Module % of development + verification + integration a year: in full for 3 years after SOP, then 50 % — rule of thumb (15–25 % a year, Edana; 21–26 % median years 1–3, 24 DACH SaaS products), not automotive-measured.
Security support after the last build	After the last build: each module's tail sustaining team × 25 % for 10 years — the duty is regulatory (UN R155: the post-production phase runs "until the end-of-life of all vehicles under the vehicle type"; R156: updates over the vehicles' lifetime); 10 years = Volkswagen / CARIAD's stated software support after end of production (2023, via the press); 25 % is a CostVision assumption — no published cost.
Homologation	Type approval ~ $1M (S&P Global: R155 on one legacy model "can easily exceed $1M"); ISO 26262 assessor £600k (ASIL-D) / £250k; organisation CSMS £1.5M + SUMS £0.8M ÷ vehicle types sharing them — CostVision estimates.
Supplier margin (bought development)	Each column re-costs the scope with that development source. A supplier's price = its effort factor ÷ (1 - 11.7 % margin), so buying is cheaper only if the supplier needs <= 88.3 % of the in-house effort (reuse, a cheaper hub mix).
Effort catalogue (person-months per module)	CostVision catalogue — no published automotive productivity source was found (the defence data in Clark & Madachy's cost-estimation manual is the next source to read). Calibrate on your own actuals or size modules in KSLOC.
Reuse, complexity factors	CostVision catalogue factors — no published source.
ASIL factor	Solcept engineering blog (× 1.4 to ASIL-B, × 1.3 B -> D); A and C interpolated — Low confidence.

Rate basis — what these figures were priced on

Rate book	CostVision built-in v2.0.0 (country book 2026-10) (reviewed 2026-10)
Engineering hub	Romania — its own salaries, statutory employer on-costs and working hours (no UK multiplier)
Hub rate (£ / person-month of 152 h, before overhead)	mid £3,359 · senior £3,359 — RON 239,265 mid / 239,265 senior gross a year + employer on-cost 2.3% at mid = 244,648 / 244,648; ÷ 1,800 productive h × 152 h = RON 20,659 / 20,659 per person-month ÷ 6.151 RON per £ = £3,359 / £3,359
Engineer rate used	£3,359 — 55 % senior blend of the hub's mid and senior rates
Development source	Tier-1 supplier × 0.997 = efficiency 0.88 (unsourced) ÷ (1 - supplier margin 11.7 %)
Engineering burden	× 1.610 = employment +15 % × company +40 % × capital +0 % (on the hub rate)
Loaded rate (£ / person-month)	£5,390
Rate confidence	Low — figures read from search results of the cited pages (Oct 2026); see the rate book
Powertrain	PHEV
Platform annual volume	not shared (this programme carries all of it)
Annual volume × life	200,000 × 6 yr
NRE recovery	6 yr
Discount rate	0 %
Development duration (tool licences)	54 months
Cloud	£10.75 per connected vehicle-year (the in-scope modules' catalogue £ / yr ÷ 440,000 reference fleet — unsourced estimate) × 100 % connected, each vehicle served 8 yr; fixed platform £0 / yr
Map / per-vehicle-year royalties	3 yr per vehicle

Development tools	£39,500 per engineer-seat-year
Maintenance profile	full 3 yr after SOP, then 50 %; wage escalation 0 % / yr
Security support after the last build	10 yr at 25 % of the tail team — After the last build: each module's tail sustaining team × 25 % for 10 years — the duty is regulatory (UN R155: the post-production phase runs "until the end-of-life of all vehicles under the vehicle type"; R156: updates over the vehicles' lifetime); 10 years = Volkswagen / CARIAD's stated software support after end of production (2023, via the press); 25 % is a CostVision assumption — no published cost.
Homologation	on — organisation CSMS / SUMS shared by 1 vehicle type(s)
Nominal effort basis	52 catalogue · 0 COCOMO II size (2.94 × KSLOC^1.0997) · 0 custom PM
Effort calibration	none (model as published)

11 Make or buy and the supplier check

Each domain's engineering re-costed with each development source at this hub. Each column re-costs the scope with that development source. A supplier's price = its effort factor ÷ (1 - 11.7 % margin), so buying is cheaper only if the supplier needs <= 88.3 % of the in-house effort (reuse, a cheaper hub mix).

Domain	OEM in-house	Tier-1 supplier	Start-up
Powertrain & Battery	£11.8M	£11.8M	£10.4M
ADAS Level 2 & 2+	£12.6M	£12.6M	£11.2M
Infotainment, Connectivity & UX	£5.8M	£5.8M	£5.1M
Vehicle Domain Controllers	£6.2M	£6.2M	£5.5M
Middleware & Platform	£2.3M	£2.3M	£2.1M
Cybersecurity (ISO 21434)	£1.3M	£1.3M	£1.2M
OTA & Cloud Backend	£1.7M	£1.7M	£1.5M

A bought source pays only when its effort is below 88.3 % of the in-house effort. The efficiency factors behind it are unsourced — set your supplier's own.

Supplier quote check

No supplier quote was entered on the Supplier check tab.

Negotiation target: a fair Tier-1 price for this scope is £45.4M of engineering at £5,390 per person-month (should-cost re-costed as a Tier-1 supplier: its effort factor ÷ (1 - 11.7 % margin)).

12 Scope, exclusions and limitations

Catalogue modules not in this programme

Module	Domain	Why not
Digital Key (UWB/BLE) & Secure Access	Infotainment, Connectivity & UX	not confirmed for this vehicle — excluded rather than assumed
48 V Belt Starter-Generator & Battery Control	Powertrain & Battery	a 48 V mild-hybrid starter-generator; not this drivetrain

Vehicle software the catalogue has no module for (review judgement)

These functions are normally supplier ECUs whose software is priced in the hardware piece price. They are not in this total unless you added them as your modules. Confirm against the electrical bill of materials.

- Occupant restraint (airbag) control
- On-board AC charger and DC-DC converter control (the charging module covers DC fast-charge protocols)
- Electric power steering and electric parking brake base software
- Tyre-pressure monitoring
- Seat memory / massage and other comfort ECUs beyond climate and body control
- Trailer functions

Limitations

- Accuracy has not been measured: no real programme cost has been logged against the model. The reconciliation in section 3 proves the arithmetic, not the inputs.
- Sourced figures (hub salaries, supplier margins, licence prices, OEM service terms) were read from search results of the cited pages, and each new default was re-checked by an independent search; the pages themselves could not be opened from the build environment.
- Published "programme software cost" figures for other vehicles are not printed: none could be verified.
- Vehicle volume and programme life are user inputs. The manufacturer publishes no model-level volume for this vehicle.
- The voice-engine royalty uses Cerence's published average price per unit as the reference price for an embedded voice licence; the vehicle's own voice-assistant terms are not public.

A Appendix — Romania engineering rate build-up

Rate confidence: Low.

RON 239,265 mid / 239,265 senior gross a year + employer on-cost 2.3% at mid = 244,648 / 244,648; ÷ 1,800 productive h × 152 h = RON 20,659 / 20,659 per person-month ÷ 6.151 RON per £ = £3,359 / £3,359

The book takes the median of the observations below: mid = median of 4 (241,042, 256,832, 237,487, 234,567) = RON 239,265; senior RON 239,265 as stated in the book.

Figure	Source	Page
Mid: RON 12,136	INS monthly earnings release, September 2025 (via economica.net) (mean NET × 1.6551 (INS 2025 whole-economy gross/net 8,716 / 5,266))	https://www.economica.net/?p=887615
Mid: RON 12,931	INS, November 2025, IT services (via hotnews.ro) (independent review) (mean NET × 1.6551 (INS 2025 whole-economy gross/net 8,716 / 5,266))	https://hotnews.ro/salariul-mediu-net-a-crescut-la-5-615-lei-in-noiembrie-2025-in-ce-domenii-s-a-castigat-cel-mai-bine-2151180
Mid: RON 11,957	INS, June 2025, IT services (via g4media.ro) (independent review) (mean NET × 1.6551 (INS 2025 whole-economy gross/net 8,716 / 5,266))	https://www.g4media.ro/castigul-salarial-mediu-net-in-luna-iunie-a-crescut-usor-la-5-539-lei-cele-mai-mari-salarii-in-industria-petroliera-si-tehnologia-informatiei-cele-mai-mici-in-horeca-si-industria-textila.html
Mid: RON 11,810	INS, October 2025, IT services (via tvrinfo.ro) (independent review) (mean NET × 1.6551 (INS 2025 whole-economy gross/net 8,716 / 5,266))	https://tvrinfo.ro/salariul-mediu-net-a-ajuns-la-5-492-de-lei-in-octombrie-it-ul-ramane-lider-la-castiguri/
Employer on-cost	work insurance contribution (CAM) 2.25% — PwC Worldwide Tax Summaries Romania; Romania puts social contributions on the employee — they are inside the gross	https://taxsummaries.pwc.com/romania/individual/other-taxes
Working hours	(260 - 20 leave - 15 public holidays) × 40 h ÷ 5 — 15 public holidays — Wikipedia list of public holidays; 20 days leave not sourced in this research — CostVision reading of the national calendar / labour code, to verify	